DAY-1

❖ **Static Routing Concepts Theory :What is static routing? Advantages and disadvantages of static routes. Practical : Configure static routes between two routers in Cisco Packet Tracer.**

1. Open Cisco Packet Tracer and set up a simple network with two routers connected via a serial cable, assigning IP addresses to both router interfaces.

ANS :

➢ What is Static Routing?

- **Static routing** is a routing method in which the network administrator manually configures the path (route) that packets should take to reach another network. Unlike dynamic routing, routers do not automatically learn or update routes.

➢ Advantages of Static Routing

1. **High Security**
   - No routing information is exchanged with other routers.
   - Less chance of unauthorized route manipulation.

2. **Low Resource Usage**
   - Uses very little CPU, RAM, and bandwidth because no routing protocol is running.

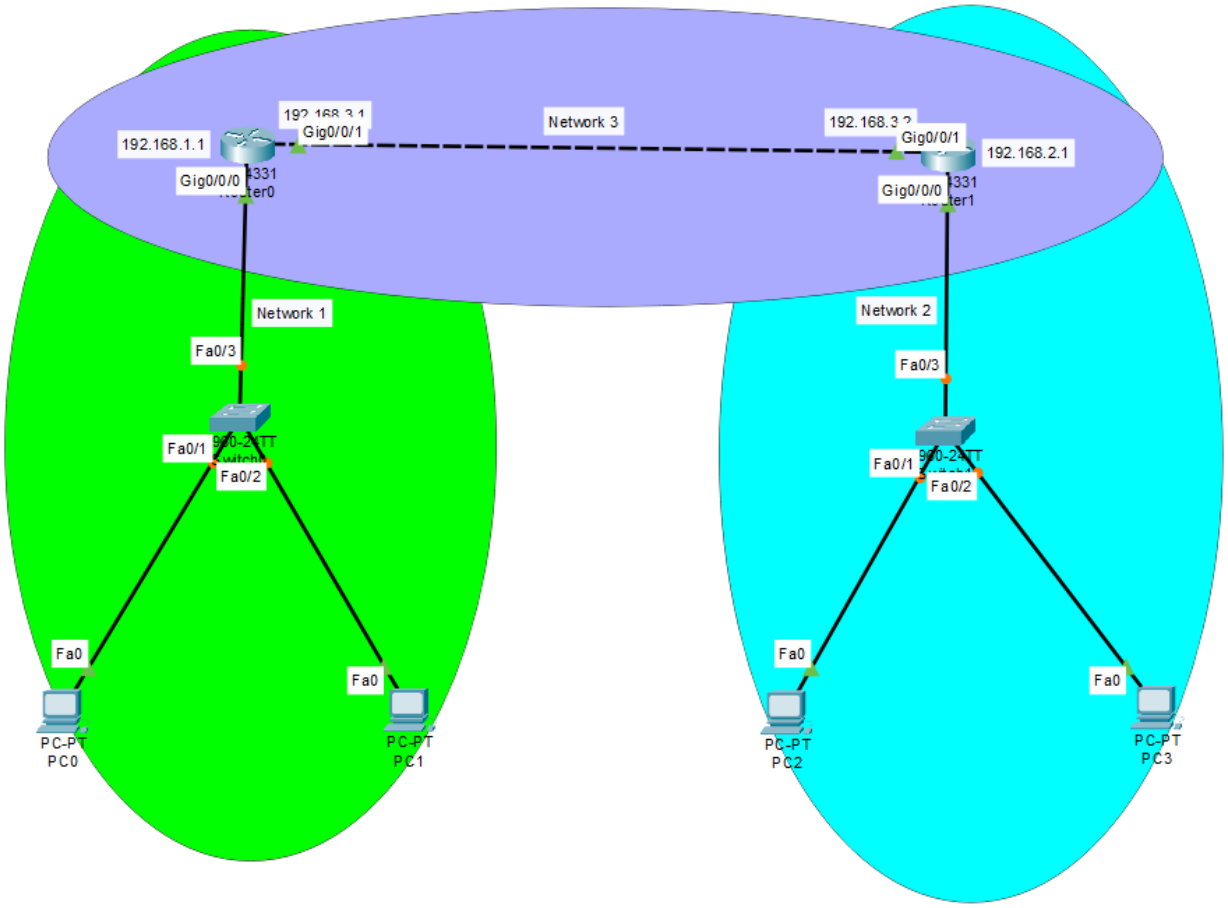
➢ Disadvantages of Static Routing

1. **Manual Configuration**
   - Every route must be configured manually.
   - Difficult to manage in large networks.

2. **No Automatic Updates**
   - If a link fails, routes are not updated automatically.
   - Administrator must manually change the routing table.

- Practical

Device	Interface	IP Address	Subnet Mask
PC1	NIC	192.168.1.2	255.255.255.0
Router1	G0/0	192.168.1.1	255.255.255.0
Router1	S0/0/0	10.0.0.1	255.255.255.252
Router2	S0/0/0	10.0.0.2	255.255.255.252
Router2	G0/0	192.168.2.1	255.255.255.0
PC2	NIC	192.168.2.2	255.255.255.0



2. List two advantages and two disadvantages of static routing compared to dynamic routing, using examples from real-life apps like IRCTC or Swiggy to illustrate your points. Hint: Think about how static routes might impact delivery or booking updates in these apps.

ANS :

- Advantages with Real-Life Examples

1. **Better Security**
   Static routes are manually configured, so unknown routers cannot exchange routing information.

   - **Example (IRCTC):**
     The connection between the booking server and the payment server can use static routing for secure communication.

2. **Low Resource Usage**
   No routing protocol runs continuously, so CPU and bandwidth usage remain low.

   - **Example (Swiggy):**
     A small local delivery office with only two branches can use static routing because the network rarely changes.

- Disadvantages with Real-Life Examples

1. **Manual Updates**
   Every network change requires manual configuration.

   - **Example (Swiggy):**
     If a new delivery hub opens, the network administrator must manually add routes to every router.

2. **Poor Scalability**
   Static routing becomes difficult when many routers are added.

   - **Example (IRCTC):**
     IRCTC has thousands of servers and branches across India. Managing all routes manually would be slow and prone to errors. Dynamic routing is more suitable for such a large network.